Product Recommendation from Transaction History

Full project report: design, implementation and results

Date: November 2026 | Status: complete | Model: Qwen3-1.7B (small language model), fine-tuned with LoRA

Summary

We built a system that reads a bank customer's recent transactions and profile, and predicts which of five products the customer is most likely to take up. The work covered four parts, all complete:

1. **Data.** A synthetic data generator, designed with a high-level and low-level design (HLD/LLD), produced 20,000 customers and 4.25 million transactions. Because the data is generated, the true chance of each customer taking each product is known.
2. **Model.** We fine-tuned an open small language model, Qwen3-1.7B, using LoRA. The model reads each customer's transactions as a sequence of compact codes, and it is trained together with a small network that reads the customer's profile.
3. **Training.** Two training rounds ran on Kaggle's free GPUs (two NVIDIA T4 cards, 16 GB each); the second took 6.4 hours. A third round trained on 50,000 customers on one NVIDIA L40S (46 GB) in 3.4 hours.
4. **Evaluation.** The model was compared with four other models on customers from later months than the training data: 5,069 in rounds 1–2 and 74,588 in round 3.

Result. Our model does not yet beat the standard LightGBM model. In round 2 (13,265 training customers) it scored 0.736 AUC, 3.8% below LightGBM (0.765). In round 3, with 50,000 training customers, it scored 0.770, 2.5% below LightGBM (0.789); the larger test set makes this gap clearly real (about ± 0.006). More data narrowed the gap, and transaction history on its own predicts take-up well above random (0.668). To beat the standard model, the next step is to first teach the model general spending patterns on all transactions (Section 9).

1 Problem and goals

Banks usually summarise each customer into a set of totals, such as "spend on groceries in the last 90 days", and train a tree model such as LightGBM on those totals. Totals are quick to compute and work well, but they lose two kinds of information:

- **Order and timing.** A pay rise followed by transfers to a broker, and late fees followed by a quick loan, can add up to the same 90-day totals.
- **Merchant detail.** Totals group shops into a few broad categories, so an airline and a bus company both become "transport".

Goal. Build a model that reads the transactions themselves, combines them with the profile, and recommends the best of five products: *travel card, personal loan, investments, insurance* and *cashback plus*. Measure it honestly against the standard approach.

Constraints. No real bank data was available, and training had to fit on free cloud GPUs (Kaggle, 2×T4, 12-hour session limit).

2 High-level design (HLD)

2.1 System overview

The system has three stages: a data generator, a training pipeline and an evaluation stage. Figure 1 shows how they connect.

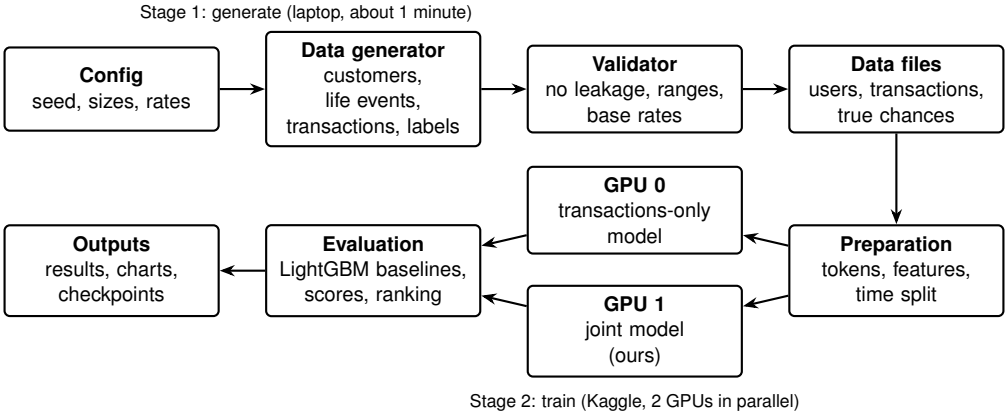


Figure 1: High-level design. Stage 1 runs on a laptop; stages 2 and 3 run as one Kaggle job.

2.2 Components

Component	Responsibility
Data generator	Creates customers with hidden traits, plants life events, simulates 150 days of transactions, computes each customer's true chance of taking each product, chooses an offer and draws the outcome.
Validator	Stops the run if any transaction is dated on or after the scoring date, if label rates fall outside the expected band, or if values are missing.
Preparation	Turns transactions into compact codes, fits amount ranges on training customers only, scales profile features, and splits customers by month.
Training workers	Two fine-tuning jobs, one per GPU: a transactions-only model and our joint model. Each keeps the checkpoint with the best validation score.
Evaluation	Trains LightGBM baselines, scores every model on the same test customers, and compares product rankings with the true chances.

2.3 Data contracts

File	Rows	Contents
`users.parquet`	20,000	One row per customer: 18 numeric and 2 categorical profile features, offered product, outcome (took it or not), split, scoring date.
`transactions.parquet`	4,251,789	Customer ID, date, signed amount, merchant description. Only dates before the customer's scoring date.
`product_truth.parquet`	100,000	True chance for every customer and every product. Used only for evaluation, never as model input.
`meta.json`	1	Generator version, seed, configuration fingerprint, row counts and base rates, for reproducibility.

2.4 Key design decisions

- **Split by time.** Customers scored in months 1–8 are used for training (13,265), month 9 for tuning (1,666) and months 10–12 for testing (5,069). This mirrors how the model would be used: trained on the past, applied to the future.
- **Known right answers.** The generator stores the true chance of each customer taking each product, so we can measure the best score any model could reach (0.867) and check whether product rankings are correct.
- **Fit on free hardware.** A small language model with LoRA adapters, half-precision maths and memory-saving training keeps each job under 7 GB of GPU memory.
- **Two models in parallel.** The two fine-tuning jobs run at the same time on the two GPUs, which halves the wall-clock time.

3 Low-level design (LLD): data generator

3.1 Modules

Module	What it does
`config.py`	One frozen settings object: 20,000 customers, seed 7, 150 days of history, 12 scoring months, 8% base rate per product, 30% targeted offers, 60% of customers with a life event.
`personas.py`	Samples seven linked hidden traits per customer (income, saver, traveller, financial stress, digital use, discretionary spend, age) and at most one life event: pay rise, financial stress, new car or new baby.
`catalog.py`	About 120 merchants in 16 categories, each with typical amount and description templates such as `UBER *TRIP {city}`.
`simulator.py`	Generates transactions: salary, rent and subscriptions on fixed days; everyday spending as random daily counts; trips and stress episodes as bursts.
`labels.py`	Computes the true chance per product, picks the offer (70% random, 30% targeted) and draws the outcome.
`features.py`	Builds the profile table: 90-day totals in 8 broad categories, noisy declared income, credit limit, app use. Deliberately loses detail, as real feature tables do.
`validate.py`, `generate.py`	Checks the data and writes the files. One command: `python -m synthgen.generate --n-users 20000 --seed 7`.

3.2 What drives each product

Each product's true chance depends on a few traits and recent events, plus random noise so that no model can be perfect. A life event only counts if it happened within the last 30–45 days.

Product	Main drivers (positive unless marked)
Travel card	Traveller trait (strong), income; financial stress (negative)
Personal loan	Financial stress; stress event in last 30 days; saver trait (negative)
Investments	Saver trait, income, pay rise in last 30 days; financial stress (negative)
Insurance	Has children; new car or baby in last 45 days; age
Cashback plus	Share of spending on restaurants, delivery and shopping; digital use

3.3 Sample data

Date	Description	Amount (R$)
2025-08-28	PAGTO SALARIO ITAU UNIBANCO	+4,872.87
2025-09-28	XP INVESTIMENTOS TED	−466.13
2025-10-28	SALARIO CONSTRUTORA ALFA	+4,912.91
2025-10-28	XP INVESTIMENTOS TED	−988.68
2025-11-08	NETFLIX.COM	−40.45
2025-11-10	PAO DE ACUCAR 3900	−338.33

Table 1: Transactions for one sample customer: salary followed by transfers to an investment broker.

4 Low-level design (LLD): model

4.1 The small language model

We chose **Qwen3-1.7B**, an open small language model (SLM). It is small enough to fine-tune on one free 16 GB GPU, it is already good at reading short text such as merchant names, and its weights are openly available.

4.2 Turning transactions into codes

Each transaction becomes six special codes followed by the first six word-pieces of the merchant name:

```
<TX> <SIGN_OUT> <AMT_7> <MON_11> <DAY_8> <WD_5> NET FL IX .COM
```

- **74 special codes**: start of transaction, money in or out, 21 amount ranges, 12 months, 31 days and 7 weekdays.

Property	Value
Parameters	about 1.7 billion
Layers	28 transformer layers
Hidden size	2,048
Attention heads	16 query heads sharing 8 key/value heads
Vocabulary	151,936 tokens
Used as	Reader only: the word-prediction layer is removed; we use the last hidden state
Precision	16-bit weights for the frozen model; 32-bit for trainable parts

Table 2: Qwen3-1.7B as used in this project.

- **Amount ranges** are fitted on training customers only, so test data never influences them.
- **The special codes have their own small table** (74 entries), started from the meaning of matching words in the base model (for example, `<MON_11>` starts from "november"). This avoids enlarging the model's 151,936-word vocabulary.
- On average a transaction takes **11.7 codes**, so the 1,024-code window holds the customer's **most recent 87 transactions**, about 9 weeks.

4.3 Model architecture

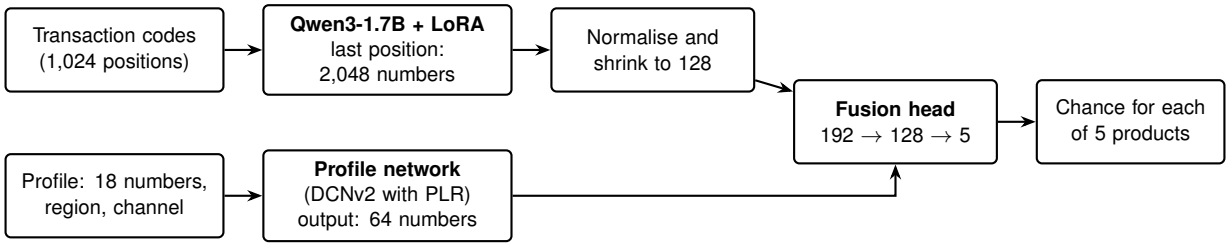


Figure 2: Our joint model. All trainable parts learn together from one loss on the offered product.

- **Profile network.** Each number is first expanded with sine and cosine features (PLR embeddings), which helps neural networks use numeric inputs. Three "cross" layers (DCNv2) then learn combinations of features, such as income together with region.
- **Fusion head.** The transaction summary (128 numbers) and the profile summary (64 numbers) are joined and passed through a small network that outputs one score per product. During training, only the score of the product actually offered is compared with the outcome.
- **Why train together.** Because one loss updates both branches, the language model is pushed to learn what the profile features miss.

4.4 Fine-tuning setup

Setting	Value and reason
Method	LoRA: small trainable add-on matrices inside the frozen model
LoRA rank / scale / dropout	16 / 32 / 0.05
LoRA placed on	All 7 projection layers (query, key, value, output, gate, up, down) in all 28 layers
Trainable parameters	18.3 million in total, about 1% of the model (17.4M LoRA + special-code table + heads)
Optimiser	AdamW; learning rate 2e-4 for LoRA, 1e-3 for the heads
Schedule	5% warm-up, then cosine decay; gradient clipping at 1.0
Batch	4 customers $\times$ 8 accumulation steps = 32 per update; 414 updates (1 pass)
Memory saving	16-bit mixed precision with loss scaling; gradient checkpointing
Model selection	Validation AUC checked 4 times; best checkpoint kept
Peak GPU memory	7.0 GB of 16 GB
Training time	4.1 hours (transactions-only), 4.3 hours (joint), in parallel

5 Code used for fine-tuning

The full training script is a single Kaggle script of about 500 lines. The excerpts below are lightly trimmed for reading.

Code 1: Building each customer's code sequence, newest transactions first until the window is full.

```
edges  = np.quantile(log_amount[train_rows], np.linspace(0, 1, 22)[1:-1])   # 21 amount ranges, train only
bucket = np.searchsorted(edges, log_amount, side="right")
head = np.stack([TX, sign, AMT_0 + bucket, MON_1 + month - 1, DAY_1 + day - 1, WD_0 + weekday], 1)

for start, end in customer_ranges:
    pieces, total = [], 0
    for i in range(end - 1, start - 1, -1):                 # walk back from the newest transaction
        p = list(head[i]) + desc_ids[descs[i]][:6]           # 6 codes + first 6 word-pieces
        if total + len(p) > 1024: break
        pieces.append(p); total += len(p)
    seqs[customer] = [t for p in reversed(pieces) for t in p]
```

Code 2: Loading Qwen3-1.7B, adding LoRA, and the separate table for the 74 special codes.

```
lm = AutoModelForCausalLM.from_pretrained("Qwen/Qwen3-1.7B", torch_dtype=torch.float16)
backbone = lm.model                                          # drop the word-prediction layer
for p in backbone.parameters(): p.requires_grad_(False)     # freeze the base model
backbone = get_peft_model(backbone, LoraConfig(r=16, lora_alpha=32, lora_dropout=0.05,
           target_modules=["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"]))
backbone.gradient_checkpointing_enable()

special = nn.Embedding(74, 2048)                             # started from matching words' meanings
special.weight.data.copy_(init_from_words)

def encode(ids, mask):
    is_special = ids >= V0
    x = torch.where(is_special[..., None], special(ids - V0), word_embeddings(ids))
    h = backbone(inputs_embeds=x, attention_mask=mask).last_hidden_state
    return h[:, -1]                                          # left padding: last position is newest
```

Code 3: The fusion head: one score per product; training uses the offered product's score.

```
class Head(nn.Module):
    def __init__(self, hidden, n_num, cat_sizes):
        super().__init__()
        self.ln   = nn.LayerNorm(hidden)
        self.proj = nn.Sequential(nn.Linear(hidden, 128), nn.ReLU())
        self.dcn  = DCNv2(n_num, cat_sizes)                  # profile network, 64 outputs
        self.mlp  = nn.Sequential(nn.Linear(192, 128), nn.ReLU(), nn.Dropout(0.1), nn.Linear(128, 5))
    def all_logits(self, h, xn, xc):
        z = torch.cat([self.proj(self.ln(h)), self.dcn(xn, xc[:, :2])], 1)
        return self.mlp(z)                                   # (batch, 5 products)
    def forward(self, h, xn, xc):
        return self.all_logits(h, xn, xc).gather(1, xc[:, 2:3]).squeeze(1)   # offered product
```

Code 4: Training loop: mixed precision, gradient accumulation, clipping, and keeping the best checkpoint.

```
opt = torch.optim.AdamW([{"params": lora_params, "lr": 2e-4},
                         {"params": head.parameters(), "lr": 1e-3}])
scaler = torch.amp.GradScaler("cuda")
for batch, ids, mask, xn, xc, y in batches(train, size=4, shuffle=True):
    with torch.autocast("cuda", dtype=torch.float16):
        h = encode(ids, mask)
    loss = F.binary_cross_entropy_with_logits(head(h, xn, xc), y) / 8
    scaler.scale(loss).backward(); micro += 1
    if micro % 8: continue                                   # accumulate 8 batches
    set_lr(step); scaler.unscale_(opt)
    torch.nn.utils.clip_grad_norm_(trainable, 1.0)
    scaler.step(opt); scaler.update(); opt.zero_grad(); step += 1
    if step % eval_every == 0:
        val_auc = auc(y_val, score(val))
        if val_auc > best: best = val_auc; save_checkpoint()
```

Code 5: Running the two fine-tuning jobs in parallel, one per GPU.

```
procs = [subprocess.Popen([python, script, "--worker", mode, "--gpu", str(g)])
         for g, mode in ((0, "seq"), (1, "joint"))]
return_codes = [p.wait() for p in procs]
report()                                                     # baselines, scores, charts
```

6 Experiments and results

6.1 Models compared

Model	What it uses
Standard model (LightGBM)	Profile features only; up to 3,000 trees with early stopping.
Two-step model	Transaction summary from the fine-tuned model, reduced to 32 numbers and added to LightGBM.
Our model (joint)	Transactions and profile, trained together end to end.
Profile details only	The profile network (DCNv2 with PLR) on its own.
Transactions only	The fine-tuned language model with a small head, no profile.
Highest possible	The true chances from the generator; the upper limit for any model.

6.2 Scores

AUC measures how often a model ranks a customer who took the product above one who did not: 0.5 is random guessing and 1.0 is perfect. All scores are on the same 5,069 test customers, of whom 12.9% took the offered product.

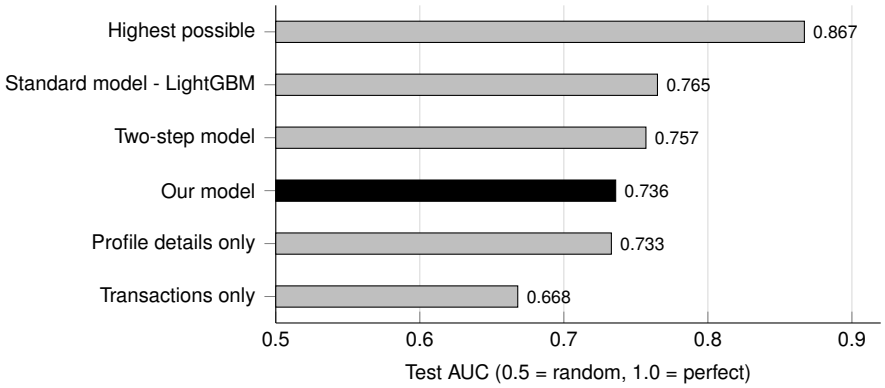


Figure 3: Test AUC of every model after round 2. Our model is the black bar.

Model	AUC	vs standard	PR-AUC	Top product right	Rank agreement
Standard model (LightGBM)	0.765	–	0.372	56%	0.63
Two-step model	0.757	−1.0%	0.379	54%	0.59
Our model (joint)	**0.736**	**−3.8%**	**0.332**	**44%**	**0.49**
Profile details only	0.733	−4.2%	–	45%	0.52
Transactions only	0.668	−12.7%	0.240	32%	0.22
Highest possible	0.867	+13.3%	–	100%	1.00

Table 3: All results. PR-AUC focuses on the customers who took the product (12.9% would be random). "Top product right" is how often the model's top product matches the true best product (20% would be random). "Rank agreement" compares each model's order of the five products with the true order (0 = no agreement, 1 = identical).

6.3 Round 1 compared with round 2

Round 1 read the last 512 codes (about 37 transactions) and treated the offered product as an input. Round 2 doubled the window to 1,024 codes (about 87 transactions), capped merchant names at 6 word-pieces, and gave each product its own output.

Model	Round 1	Round 2	Change (95% range from resampling the test set)
Transactions only	0.623	0.668	+0.045 (+0.029 to +0.062): a real improvement
Our model (joint)	0.736	0.736	+0.0005 (−0.009 to +0.010): no change

Table 4: Round 1 vs round 2 on the same test customers, with a paired bootstrap of 2,000 resamples.

6.4 Round 3: 50,000 training customers on an L40S

Round 3 tests the first two suggestions from round 2: more customers and a larger test set. We generated 300,000 customers with the same generator (seed 7, 63.8 million transactions) and used 50,000 for training, 5,000 for tuning

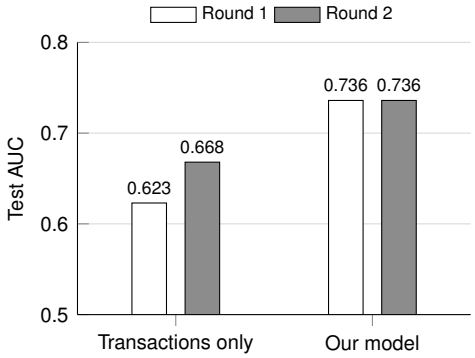


Figure 4: Test AUC in each round.

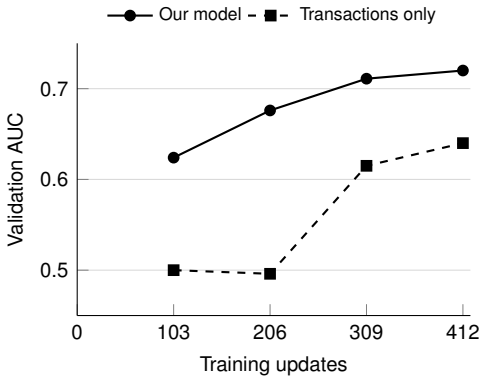


Figure 5: Validation AUC during round 2 training.

and all 74,588 customers in months 10–12 for testing (12.3% took the offered product). The model and settings were unchanged from round 2: a 1,024-code window, LoRA rank 16, one pass. The run used one NVIDIA L40S (46 GB), which removed the need for gradient checkpointing. Only the joint model and the profile-only baselines were trained, so the two-step and transactions-only models have no round 3 score.

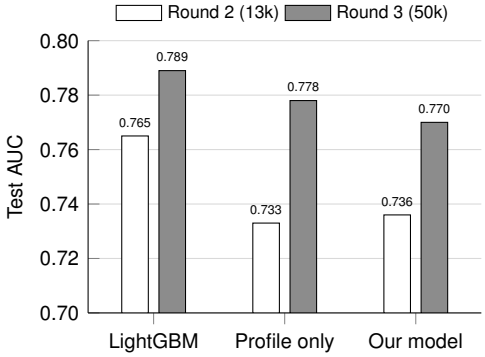


Figure 6: Test AUC with 13,265 vs 50,000 training customers.

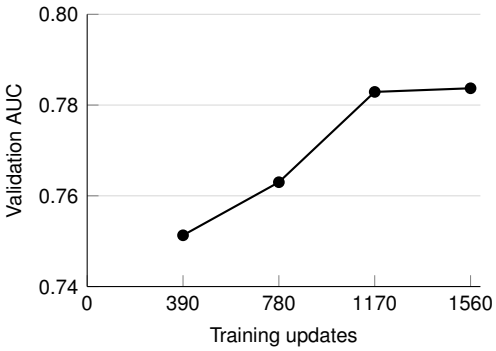


Figure 7: Validation AUC of our model during round 3; it levels off after about 1,200 updates.

Model (round 3)	AUC	vs standard	PR-AUC	Top product right	Rank agreement
Standard model (LightGBM)	0.789	–	0.390	58%	0.66
Profile details only	0.778	−1.5%	–	55%	0.63
Our model (joint)	**0.770**	**−2.5%**	**0.367**	**51%**	**0.60**
Highest possible	0.872	+10.5%	–	100%	1.00

Table 5: Round 3 results on 74,588 test customers (12.3% would be random for PR-AUC). With this many customers, each AUC is accurate to about ±0.006.

Round 3 run	Value
Hardware	1×NVIDIA L40S, 46 GB; peak memory 37.5 GB
Training	1,562 updates (batch 8 × 4 accumulation), 18.3M trainable parameters, 132 minutes
Scoring the test set	70 minutes for 74,588 customers × 5 products
Total run time	204 minutes, against 386 minutes for round 2 on 2×T4

Two cautions apply. Round 3 uses a different and larger test set, so its scores are not directly comparable customer by customer with round 2; the comparison is between the gaps to LightGBM within each round. LightGBM also improved with more data (0.765 to 0.789), which shows that part of every model's gain comes from the extra training customers rather than from the transactions.

6.5 What the results show

- **Transaction history carries real signal.** On its own it reaches 0.668, and reading more history improved it clearly.
- **Our model does not yet beat the standard model.** It was 3.8% below LightGBM in round 2 and 2.5% below in round 3. With 74,588 test customers the round 3 gap (0.019) is about three times the measurement error, so it is real.
- **More data helps our model more than it helps LightGBM.** Our model gained 0.034 AUC from 4× more training

customers; LightGBM gained 0.024. The gap closed by about a third.
- **In round 3 the transactions did not add to the profile.** Our model scored below the profile network alone (0.770 vs 0.778). On this synthetic data, the profile's 90-day totals already carry most of the signal the transactions hold, and one pass of fine-tuning is not enough for the language model to add more.
- **Validation scores levelled off** in round 3 (0.783 to 0.784 over the last 390 updates), so simply training longer on the same setup is unlikely to close the gap.

7 Problems found and how we solved them

Problem	What we did	Result
No real bank data	Built a generator that also stores each customer's true chance per product	Highest possible score is known
Transactions as text are long; amounts read poorly	Six compact codes per transaction plus a short merchant name	11.7 codes per transaction
Enlarging the vocabulary would add 311M parameters	Separate 74-entry table for the special codes	Base vocabulary unchanged
Model too large to fully train on free GPUs	LoRA (about 1% trainable), 16-bit maths, gradient checkpointing	7 GB peak memory
Risk of testing on seen data	Split by month; amount ranges and scaling fitted on training data only	Fair test on future customers
Library conflict on Kaggle (PEFT and torchao)	Disabled the unused torchao check at start-up	Training runs cleanly
Round 1 saw only 37 transactions	Doubled the window; capped merchant names at 6 word-pieces	Transactions-only AUC 0.623 $\rightarrow$ 0.668
Round 1 ranked products worse than chance	One output per product instead of the offer as an input	Rank agreement $-0.22 \rightarrow +0.22$

8 Limitations

- **One training pass.** Rounds 1–2 used 13,265 customers (414 updates) and round 3 used 50,000 (1,562 updates), each for a single pass.
- **No spending-pattern pre-training.** Qwen3 learned from general web text; it has never been taught how bank transactions usually follow each other.
- **Synthetic data.** The generator encodes our assumptions about behaviour. Real data will differ, and in this generator the 90-day totals already capture much of the signal.
- **One outcome per customer.** We only see whether the customer took the product they were offered, not the other four.
- **Two changes in round 2 at once.** The window size and the output head changed together, so their separate effects are unknown.

9 Suggested steps to improve the results

1. **Teach the model spending patterns first.** Train it to predict the next transaction across all 4.25 million transactions (no labels needed), then fine-tune it for recommendation. This is expected to give the largest gain.
2. **Use more customers (done in round 3).** 50,000 training customers narrowed the gap from 3.8% to 2.5% but did not close it. Scaling further, to 200,000+, is worth one more run; extra passes on the same data are less promising because validation scores levelled off.
3. **Use a larger test set (done in round 3).** 74,588 test customers give about ± 0.006 accuracy, enough to measure a 1% improvement.
4. **Change one thing at a time.** Run separate rounds for window size, head design and learning rate, so each effect is known.
5. **Use a larger GPU (done in round 3).** One L40S ran 4$\times$ more training data in 3.4 hours, against 6.4 hours for round 2 on Kaggle's 2$\times$T4.

6. **Try a longer window and a stronger profile model.** A 2,048-code window would cover about 4 months; adding LightGBM's predictions as an input to the fusion head may also help.
7. **Test on real, anonymised bank data** to confirm the results hold outside the synthetic setting.

Appendix: how to reproduce

1. Generate the data: `python -m synthgen.generate --n-users 20000 --seed 7 --out data/`
2. Upload `data/` as a Kaggle dataset and attach the Qwen3-1.7B model.
3. Run the training script as a Kaggle notebook with 2×T4 GPUs. It prepares the data, runs both fine-tuning jobs in parallel, trains the baselines and writes `results.json` and the charts.
4. For a quick local test without a GPU: run the training script with `--smoke`.

Total run time for round 2: 386 minutes. All random seeds are fixed (generator 7, training 42).

Round 3: generate 300,000 customers with `python -m synthgen.generate_big --n-users 300000 --seed 7 --out data_300k`, then run the training script on one GPU with `DATA=data_300k N_TRAIN=50000 N_VAL=5000 MODES=joint`. Total run time: 204 minutes.